

Justin Chen

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Education

University of Maryland

Expected May 2027

Double Major: Computer Science - Machine Learning Track and Business Analytics (GPA: 3.8 / 4.0)

College Park, MD

- **Awards:** Four-Year UMD President's Scholarship (\$40,000), HackTJ Winner 2023 (300+ participants), HooHacks 3rd Place 2024 (750+ participants), Bitcamp Winner 2024 (600+ participants), Appian Hackathon Winner 2025

Experience

Amazon

May 2026 – Aug 2026

Software Engineer Intern

Sunnyvale, CA

- Designed and implemented an **end-to-end**, in-route **search feature** for **Amazon Smart Vehicles** (BMW integration)
- Scaled Alexa+ **LLM** capabilities by redesigning **API schemas** and instruction prompts, achieving a $\geq 95\%$ accuracy rate in inferring user intent during **natural language processing**.
- Optimized **geospatial** backend logic from **live navigation streams**, ensuring high search performance (**p99** $\leq 500\text{ms}$)
- Shipped a feature using, **C++**, **Java**, and **Kotlin** for drivers to query points of interest using time and distance offsets

Polar Dream Travel

January 2026 – May 2026

Software Engineer Intern

Mclean, VA

- Architected and deployed the **technical infrastructure** for a boutique travel agency managing **400k** annual revenue
- Developed a (**RAG**) **AI email agent** utilizing **OpenAI embeddings** to parse incoming customer inquiries, achieving a **92%** automated resolution accuracy rate.
- Designed a **PostgreSQL** database schema using **Supabase**, implementing row-level security (**RLS**) to securely store **10k+** customer profiles and transactional cruise ticketing records.

Appian

June 2025 – Aug 2025

Software Engineer Intern

Mclean, VA

- Deployed and maintained production-grade **Kubernetes** clusters in **AWS** using **Python**, **Terraform**, **Go** and **Helm**
- Improved network architecture for **distributed systems** with **Kubernetes** and **Terraform**, cutting server load by **18%**
- Led an update to critical **Karpenter** operator features, making server balancing more efficient and reducing costs by **\$11,000** annually
- Developed and **fine-tuned** a Retrieval-Augmented Generation (**RAG**) **AI agent** leveraging a custom knowledge base to interpret code and automatically generate context-aware documentation updates, reducing update time by **60%**
- Enabled **AWS S3** native state-locking by updating **DynamoDB tables** and **S3 buckets**, cutting costs by **10%**

General Dynamics Mission Systems

May 2024 – Aug 2024

Software Engineer Intern

Manassas, VA

- Engineered and sustained **Java**, **C++**, and **Python** software for US Navy **data collection** and transmission systems
- Designed a **Bash** script to verify media (drive partitioning, encryption, filesystems) on **thousands** of drives to test Johns Hopkins research data.
- Upgraded software to utilize **Java NIO** for **file I/O operations** on digital records, resulting in a **50%** speed increase
- Performed Fortify **static code analysis** to identify and implement fixes for **30+** identified code weaknesses in C++

Projects

Machine Learning Color Detection Program | Python, NumPy, OpenCV, Sckit-learn

- Architected a **machine learning** driven color detection system, employing Gaussian Mixture Models trained on a diverse dataset of over **100 images**, optimizing **15** critical model parameters

AI Stock Market Pattern Analysis Trading Bot | Python, SQL, PyTorch, Pyplot

- Leveraged **AI** and **pattern recognition algorithms**, coupled with **SQL** database querying, to perform comprehensive analysis of stock market data. Used **data visualization** graphs to provide users with user-friendly data analysis

Technical Skills

Languages: Python, Java, C, C++, Bash, Go, SQL, JavaScript, HTML, CSS, Kotlin, OCaml, Rust, R, MATLAB

Technologies: Kubernetes, Terraform, AWS, Docker, React, Flask, Unix/Linux, Git, Tableau, Excel

Relevant Coursework: Artificial Intelligence, Machine Learning, Algorithms, Object-Oriented Programming, Computer Vision, Data Science, Android/iOS App Development, Operating Systems, Database Algorithms